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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Fifth Semester B.Tech (CBCS) Degree Examinations

December 2025

CV2109-1 – DESIGN AND CONSTRUCTION OF HIGHWAYS AND PAVEMENTS

(For CIV)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: i) Answer **Five full** questions choosing **Two full** question from each **Unit**.

ii) The usage of code books IRC 37, IRC 58, and IRC 115 shall be permitted

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which road network pattern involves radial roads radiating outward from the Central Business Area (CBA) to the outskirts?
A) Rectangular or block pattern
B) Star and block pattern
C) Radial or star and circular pattern
D) Star and grid pattern
2. Ideal shape of the vertical summit curve in road is
A) Cubic parabola
B) Circular parabola
C) Compound parabola
D) Combined parabola
3. Design of horizontal and vertical alignments, super-elevation, sight distance and grades, is worse affected by _____
A) Width of the vehicle
B) Length of the vehicle
C) Height of the vehicle
D) Speed of the vehicle
4. The important factor considered in the design of summit curve on highways is
A) Comfort to passengers
B) Sight distance
C) Super elevation
D) Impact factor
5. In the plasticity chart, where do silt and organic soils exist in relation to the A-line?
A) Above the A-line
B) Below the A-line
C) A-line has no influence
D) Exactly on the A-line
6. In crushing test, dry aggregates passing through _____ mm sieve and retained _____ mm in a cylinder
A) 12.5, 10
B) 10, 2.36
C) 20, 16
D) 20, 4.75
7. Which of the following gradation has more fraction of coarse aggregate and less fraction of fine aggregate?
A) Uniform
B) Dense
C) Open
D) Gap
8. Kinematic viscosity is related to absolute viscosity by _____
A) Volume
B) Mass
C) Weight
D) Density
9. For design of rigid pavement, the modulus of elasticity and Poisson's ratio of concrete are _____ & _____
A) 30000 MPa, 0.15
B) 3000 MPa, 0.2
C) 3000 MPa, 0.25
D) 30000 MPa, 0.2
10. In flexible pavement, the load transfers from top layer to bottom layer through _____
A) Slab action
B) Beam action
C) Grain to grain contact
D) Capillary action



11. What is the structural analysis of flexible pavement using IIT Pave based on?
 - A) Elastic theory
 - B) Multi-layer theory
 - C) Linear theory
 - D) Fatigue theory
12. The vehicles which have _____ spacing between front axle and first rear axle, are critical for Top Down Cracking (TDC).
 - A) Less than 4.5 m
 - B) Equal to 4.5 m
 - C) Greater than 4.5 m
 - D) Less than 3.5 m
13. _____ spread the concrete equally and uniformly in front of the metering gate of slipform paver
 - A) Super Smoother
 - B) Mould unit
 - C) Plough unit
 - D) Oscillating beam
14. A _____ roller consists of a flat platform with number of smooth rubber tyre wheels mounted on two or more axles
 - A) Sheep's foot
 - B) Smooth wheeled
 - C) Dead weight
 - D) Pneumatic
15. _____ helps in planning and levelling the resources as per the availability and requirement of the project.
 - A) Baseline program
 - B) Resource histogram
 - C) Strip chart
 - D) Gantt chart
16. _____ is a level of planning involving further division of the project work into major components
 - A) Project milestone schedule
 - B) Resource loading
 - C) Cost loading
 - D) Work breakdown structure
17. What is used to reinforce the subbase or slope according to the specifications in MoRTH clause 703?
 - A) Prime coat
 - B) Elastomeric rubber membrane
 - C) Geogrids
 - D) Tack coat
18. What is the purpose of a separation membrane in the PQC layer?
 - A) Enhance water retention in immature PQC
 - B) Promote crack propagation
 - C) Increase frictional forces
 - D) Act as a joint sealing compound
19. In cement concrete slabs in pavements, tie bars are installed in
 - A) Expansion joints
 - B) Longitudinal joints
 - C) Warping joints
 - D) Contraction joints
20. For deflection measurement through Falling weight deflectometer testing, which type of load is applied?
 - A) Static Load
 - B) Moving Load
 - C) Continuous applying load
 - D) Impact Load

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*												
1.	a) Explain the different types of sight distances in highway design. For each type, describe its definition, purpose, and factors affecting it. Include the relevant formulas where applicable.	8	L*3	1	1												
	b) Match the items in Column A with their corresponding keywords in Column B. For each match, describe why you selected that option.																
<table><tr><th>Column A</th><th>Column B</th></tr><tr><td>1. Penetration Test</td><td>A. Consistency</td></tr><tr><td>2. Ductility Test</td><td>B. Elasticity</td></tr><tr><td>3. Flash Point Test</td><td>C. Ignition</td></tr><tr><td>4. Kinematic Viscosity Test</td><td>D. Fluidity</td></tr><tr><td>5. Loss on Heating Test</td><td>E. Stability</td></tr></table>		Column A	Column B	1. Penetration Test	A. Consistency	2. Ductility Test	B. Elasticity	3. Flash Point Test	C. Ignition	4. Kinematic Viscosity Test	D. Fluidity	5. Loss on Heating Test	E. Stability	4	L4	2	2
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c)	Evaluate how insufficient drainage affects road safety and durability. What effective solutions can be recommended to mitigate these problems?	4	L5	2	4
2. a)	Outline the steps involved in highway geometry design using Civil 3D.	4	L3	1	1
b)	In a city's business district, major roads radiate outward from a central business area, with smaller streets arranged in a block layout along these radials to reduce congestion while maintaining access to the center.				
i)	Identify this road pattern.				
ii)	Draw a diagram of the pattern.				
	Explain its advantages and disadvantages	4	L5	1	3
c)	Assess the effectiveness of various highway drainage systems in addressing rainfall-related challenges.	8	L4	2	4
3. a)	Describe the key components of a highway cross-section and their roles in ensuring safety and functionality. How do these elements interact to enhance the driving experience overall?	8	L3	1	1
b)	What insights does the hydrometer analysis test offer about soil characteristics and behavior and explain steps are involved in performing this assessment?	8	L4	2	4
Unit – II					
4. a)	Explain the necessary input parameters for a three-layer flexible pavement section in IITPAVE software as per IRC: 37 guidelines. Include a labeled sketch of the pavement section in your response. Additionally, describe IITPAVE's output parameters and their role in calculating various design-related parameters.	8	L5	3	5
b)	Explain the operation and components of paving equipment and provide a detailed explanation of the procedure for laying.	8	L3	4	5
5. a)	An embankment has a California Bearing Ratio of 7%, while the CBR of the borrow soil used for the subgrade is 14% for a flexible pavement design. Calculate the effective subgrade modulus of the embankment and the resilient modulus of the Granular Sub-Base. The deflection measured under the pavement is 2.4 mm, and the thicknesses of the Wet Mix Macadam and GSB layers are 250 mm and 200 mm, respectively.	8	L3	3	1
b)	Detail the steps involved in the operation of an asphalt batch mixing plant.	4	L3	4	1
c)	Explain the different levels involved in the planning of a road project	4	L3	4	2
6. a)	Explain the critical stresses and the critical positions of loading in a Pavement Quality Concrete slab.	4	L3	3	4
b)	Based on IRC : 37-2018 guidelines and the provided data of a design life of 15 years, a lane distribution factor of 0.75, an annual growth rate of commercial vehicles at 6%, a vehicle damage factor of 4, and an initial traffic volume of 3000 Commercial Vehicles Per Day (CVPD) upon completion of construction, what is the design traffic in terms of cumulative number of standard axles (in million standard axles, up to two decimal places)?	4	L5	3	2

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| c) How do project managers track and monitor the development of highway projects effectively? | 8 | L3 | 4 | 2 |
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Unit – III

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| 7. a) Describe the Construction Practices and Quality Control followed in construction of Pavement Quality Concrete layer. | 8 | L3 | 5 | 3 |
| b) Explain the different types of defects in rigid pavement that can be observed and discuss how these defects impact road safety and performance. | 8 | L3 | 5 | 2 |
| 8. a) Discuss the significance and types of joints in rigid pavements and describe how they contribute to controlling stresses and improving pavement performance. | 8 | L4 | 5 | 4 |
| b) How do prime and tack coats differ in their roles within flexible pavement construction? | 4 | L4 | 5 | 1 |
| c) Why is functional evaluation important for maintaining pavement safety and quality for road users? | 4 | L3 | 5 | 2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome.
